Queen Mary University of London

EBU5608 Product Development and Management

Topic 4 – Innovation

1

Queen Mary University of London

Agenda

- Innovation and product development
- Types of innovation
- Models of innovation
- What makes an innovative company?

2

Innovation and product development

- Innovation originates from Latin „**Innovare**" = meaning to **renew**, to make **new** or to alter, new way of doing things.
- Innovation in the context of product development refers to the process of creating new or significantly improved products, services, or solutions that address unmet customer needs, solve problems, or enhance existing offerings.
- It involves the application of creative ideas, new technologies, or novel approaches to improve the functionality, design, user experience, or overall value proposition of a product.
- Innovation drives the creation of competitive products that can disrupt markets, improve customer satisfaction, and create value for both the company and the customer.

3

Innovation and product development

- Key aspects of innovation in product development include:
  - **Novelty**: Introducing new features, functions, or designs that haven't been seen before or taking existing products and improving them in a meaningful way.
  - **Market Relevance**: Ensuring that the innovation meets the needs, desires, or pain points of the target audience.
  - **Commercial Viability**: The product needs to be feasible in terms of production, cost, and scalability to reach the market effectively.

4

Queen Mary University of London

Innovation

- **Definition**: Innovation involves improving or making significant advancements to an existing product, service, or process. It can also refer to introducing a new idea or method to the market in a way that adds value.
- **Focus**: The focus is on the **implementation** of an invention or an improvement that brings **practical, valuable, and marketable** outcomes.
- **Outcome**: Innovation makes an invention or idea **useful** or accessible to a broader audience, often making it more functional, efficient, or scalable.
- **Example**: The **smartphone** is an innovation based on the invention of the telephone, integrating various technologies (camera, internet, GPS, etc.) into one device that changes the way we live and interact.

5

Queen Mary University of London

Invention

- **Definition**: Invention is the creation of something entirely new—an original idea, concept, or device that has never existed before.
- **Focus**: The focus is on the **novelty** of the product or idea.
- **Outcome**: Inventions typically result in a **new product, technology, or process** that addresses a specific problem or fulfills a particular need.
- **Example**: The **lightbulb** invented by Thomas Edison or the **telephone** invented by Alexander Graham Bell. These were groundbreaking ideas that did not exist before.

6

Queen Mary University of London

Invention and Innovation – What is the difference?

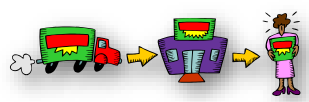
- **Creation vs. Improvement**: Invention is the creation of something new, while innovation is the improvement or application of something existing to make it more useful or valuable.
- **Novelty vs. Practicality**: An invention is novel, whereas innovation focuses on the practical application and market adaptation of an idea.
- **Scope**: An invention can be a component or idea that is the foundation of future innovations. Innovation often involves both the application of new ideas and the refinement of existing ones to meet real-world needs.
- **Invention** brings new things into existence, while **innovation** makes those things useful and effective in the real world.

7

Queen Mary University of London

Types of innovation

- We can categorise innovation in different ways:
  - **Product** (or service) v. **Process** innovation
  - **Incremental** v. **Radical** innovation
- Further types are explored in the table on the next slides



9

Queen Mary University of London

Types of innovation

Product Innovation	The development of a new or improved product
Process innovation	The development of a new manufacturing process
Organisational innovation	A new venture division, a new internal communication system, introduction of a new accounting procedure
Management innovation	TQM (Total Quality Management) systems. BPR (business process re-engineering)
Production innovation	Quality circles, just-in-time (JIT) manufacturing system, new production planning software
Commercial/marketing innovation	New financing arrangements, new sales approach e.g. direct marketing, e-commerce
Service innovation	Telephone financial services or on-line services e.g. customer support

10

Queen Mary University of London

Product Innovation

- In most cases we think of innovation relating to **actual products**
  - New technologies
  - Leading edge products
  - Gadgets, etc
- **Advantages** of launching new or improved products on to the market include [2]:
  - Increased market share
  - Public relations – e.g. news coverage
  - Enhanced reputation as an innovative company
  - Opportunity to build early customer loyalty
  - Added value
  - Higher prices and profitability
  - Competitive advantage

However, creation of some innovative products can instigate **other innovations**

11

Queen Mary University of London

Product Innovation

- **Product** innovation can lead to **process**, **management** and/or **organisational** innovation
- The **digital camera** replacing film cameras, eg:
  - **Changes** had to take place in manufacturing, marketing, sales and repair functions
  - **Business focus** shifted to digital cameras for the consumer market
  - **Marketing staff** were required for promotion and education of customers (and distributors)
  - **Production** of the cameras is different – new processes, equipment and human skills
  - **Complementary goods** are different (i.e. memory cards instead of film)

12

Queen Mary University of London

Process Innovation

- Has to do with finding more **efficient** and **effective** ways of
  - **Producing** existing products
  - **Delivering** existing products
  - Can also include producing or delivering **new** products were appropriate
- **Advantages** are
  - Reduced **costs**
  - Improved **quality**
  - More responsive **customer service**
  - Greater **flexibility**
- **Disadvantages**
  - Loss of **jobs** – especially if work is outsourced
  - Need for **re-training** of workers

13

Queen Mary University of London

Models of innovation – What is the source?

- **Serendipity**
  - Innovation luck? *'Eureka!'*

- Linear models (3 models)
  1. **Technology push**

R&D → Manufacturing → Marketing → User

- Scientists make unexpected discoveries
- Technologists apply them to develop product ideas
- Engineers and Designers turn them into prototypes for testing
- Manufacturing devise ways of producing the products efficiently
- Marketing and sales will promote the product to the potential customer

[1]

14

Queen Mary University of London

Models of innovation – Market pull

- Linear Models – 2. **Market pull**

Marketing → R&D → Manufacturing → User

- Identified in 1970's,
- Studies showed that market was influential in the innovation process,
- This is a more customer driven model and is used by the majority of industries today,
- Market Research supports this model,
- Customer requirements are then conveyed to R&D for design and engineering and then manufacturing for production.

[5]

15

Queen Mary University of London

Models of innovation – University & industry liaison

- Linear Models – 3. **Universities & industry liaison**

Creation of **new knowledge**, dominated by universities and large science-based organisations

Technology development, dominated by organisations

Consumers express their **needs and wants** through the consumption of products

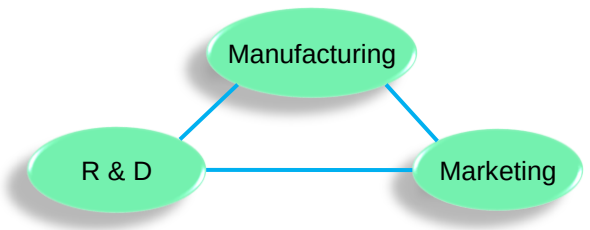
[5]

16

Queen Mary University of London

Models of innovation – Simultaneous coupling

- **Simultaneous coupling**



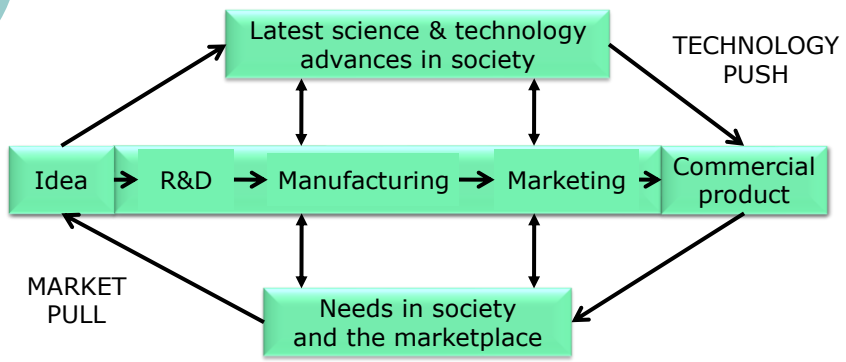
- Identified in the 1980's,
- Within this model, the moment of innovation is unknown (unlike the linear models),
- The groups work together simultaneously and create innovation.

17

Queen Mary University of London

Models of innovation – Interactive

- **Interactive**
  - Innovations occur due to work in universities, industry and from market demand



18

Queen Mary University of London

Models of innovation – Interactive (cont.)

- This model takes **simultaneous** coupling one step further
- No explicit **starting** point
- Identified in 1980's/1990's by Rothwell and Zegveld
  - They believed that society had an influence on innovation and vice versa [10]
- The diagram represents the links between the **organisation** and its internal and external linkages with the **marketplace** and **science** base
- Organisations which can **successfully** manage this process will be the most **successful innovators**
- This diagram allows for **feedback** through the different stages of the development of innovations.

19

Queen Mary University of London

What makes a company innovative?

Organisational requirement	Characterised by
Growth orientation	A commitment to long-term growth rather than short-term profit
Vigilance	The ability of the organisation to be aware of its threats and opportunities
Commitment to **technology**	The willingness to invest in the long-term development of technology
Acceptance of **risks**	The willingness to include risky opportunities in a balanced portfolio
Cross-functional **cooperation**	Mutual respect among individuals and a willingness to work together across functions
Receptivity	The ability to be aware of, to identify and to take effective advantage of externally developed technology
'slack'	An ability to manage the innovation dilemma and provide room for creativity
Adaptability	A readiness to accept change
Diverse range of **skills**	A combination of specialisation and diversity of knowledge and skills

21

Queen Mary University of London

What makes a company innovative – Investment

- Growth orientation
  - **Not all** companies' priority is for growth
  - Often **family run** companies want to stay the same size, thus remaining manageable without external intervention
  - **Innovative** companies are those that wish to grow their business
  - This is a **long-term** process, and investment is required
  - Companies show this R&D investment in their **company reports**
    - E.g. Amazon, Microsoft, Siemens, Tencent, others?

22

Queen Mary University of London

What makes a company innovative – Vigilance

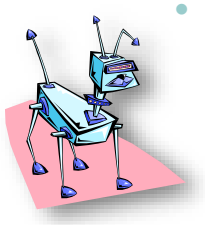
- It is vital to be aware of what markets and competitors are doing
- **SWOT** analysis is an important tool
- Identifying **threats**, **opportunities** etc.
- Often carried out by **marketing** team, but should be encouraged in **all departments** at all levels
- **R&D** team University liaison
- Many companies have specific teams to **monitor** competition and opportunities

23

Queen Mary University of London

What makes a company innovative – Commitment to technology

- **Long-term** commitment is needed, and appropriate resources must be financed
- There is not always an immediate return on **investment**
- Some companies **moved away** from this during the IT/Telecoms recession
- **Creativity** is only seen in a climate of commitment to innovation
  - Need to employ and retain innovative **staff**



24

Queen Mary University of London

What makes a company innovative – Acceptance of risks

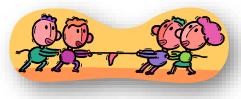
- Not all investment in innovation will be realised in **turnover** or **profit**
- Not all inventions will become **innovations**
- **Risk assessment** strategies and balanced product portfolios are crucial to success
- The business **strategy** must consider risk and commitment

25

Queen Mary University of London

What makes a company innovative – Cross-functional cooperation

- Reduction/removal of inter-departmental **conflict**
- The biggest conflict is often between **marketing** and **R&D** functions
  - Opinions differ over business **objectives** and the ability to commit to firm **dates**
  - The marketing function does not always understand what is involved **technically** to create product



26

What makes a company innovative – Receptivity

- The ability to accept technologies from **outside** of the company and use them to create new innovations
- It is very difficult to create everything 'in-house'
- Strength to identify areas of **focus**
- **Joint activities** allow knowledge sharing
  - e.g. IBM, Microsoft joint activity to create mutually beneficial technology, including web services and creating industry standards

27

What makes a company innovative – Slack

- Allowing individuals time to **think** and investigate new technologies etc.
- Some **flexibility** on milestones
- **Encouragement** for group discussions, attending conferences etc.

28

Queen Mary University of London

What makes a company innovative – Adaptability

- Ability for companies to **adapt** as required
- Innovative products can lead to **changes** in process etc.
- Need to encourage a **climate** that accepts change
- Can be easier in **smaller companies** with less processes.
  - From its origins in explosives, DuPont added other businesses like lacquers and synthetic rubber before inventing the first polyesters, nylon, Teflon, and the first phenothiazine insecticide.

29

Queen Mary University of London

What makes a company innovative – Diverse range of skills

- Ensuring that a company has a **work force** with a diverse range of skills
- Important to have employees with the appropriate **industry knowledge**
- Good **management** knowledge and experience
- Strong overall team with the correct **mix** of skills and experience
- Innovation ecosystems rely on a wide range of skills, such as technicians, engineers, discovery scientists, technologists, entrepreneurs, and investors.

30

Queen Mary University of London

Summary

- Innovation and product development
- Types of innovation
- Models of innovation
- What makes a company innovative?

31

Queen Mary University of London

Reading

- Deborah. L. Roberts, Roger Palmer and Mathew Hughes (2021), *Innovating the product innovation process to enable co- creation*, RADMA and John Wiley & Sons Ltd.

  pages 4 - 32

34

Queen Mary
University of London

References

[1] UK Innovation Strategy Leading the future by creating it, Department for Business, Energy & Industrial Strategy, July 2021, page 17

35